

## System Architecture

The architecture diagram presents the end-to-end processing flow of the implemented phishing email detection system. Email text is obtained from an open Gmail message or entered manually through the Chrome extension. The extension sends the text as a JSON request to the Flask /api/analyse endpoint. After request validation, the backend normalises the email, tokenises the text and calculates twelve engineered NLP features. These inputs are processed by the hybrid DeBERTa model to produce a phishing or legitimate verdict and a model score. LIME then creates a local explanation by identifying words that supported or opposed the displayed prediction. Flask returns the verdict, score and explanation as JSON, which the extension presents in a readable format.

